

Assignment 3.

Red

Q1)

- A solar collector is a device that captures solar radiation and converts it into thermal energy (heat). This heat is transferred to a working fluid like water, Air, Oil. They are used in water heating, space heating, Industrial heat applications.

Working Principle of Solar collector.

- 1) Sunlight falls on the collector surface.
- 2) Absorber plate absorbs radiation
- 3) Heat is transferred to fluid flowing through tubes.
- 4) Hot fluid is stored or used for heating/power generation.

Q2)

Types of Solar Collectors.

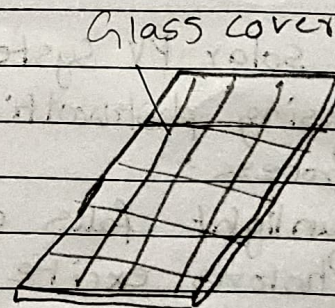
1) Non-Concentrating Collectors

1) Flat plate collector (FPC)

- Most common type of solar collector
- Used for domestic hot water.
- Glass cover, Black absorber plate.

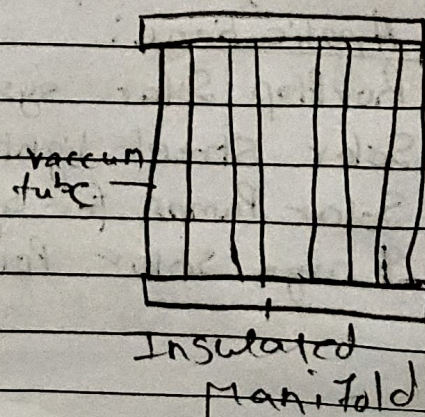
Tubes for water circulation, insulated box.

- Application: Solar water heater, Space heating.



2) Evacuated tube collector

- Higher efficiency
- Works well in cold climates
- Vacuum tubes reduce heat loss
- Water gets heated quickly
- App:- Hotels, hospitals, industries.



B) Concentrating Collectors

1) Parabolic ~~Collector~~ Trough Collector

- Cylindrical mirror focuses sunlight onto absorber tube.
- Solar thermal power plant & Industrial steam generation are its applications.

2) Parabolic Dish collector

- Dish mirror focuses sunlight at one point receiver
- High temperature power generation

3) Central Tower (Heliostat collector)

- Many mirror reflect sunlight onto a central receiver
- Large CSP plant worldwide.

A 2)

→ A solar PV system converts sunlight directly into electricity using photovoltaic effect process.

- 1) Sunlight falls on PV cells.
- 2) Photons excite electrons in semiconductor material.
- 3) This creates DC electricity.
- 4) Inverter converts DC $\rightarrow$ AC power.
- 5) Power is used for loads or fed to the grid.
- 6) Battery storage can store extra energy.

Applications:

- 1) Rooftop Solar systems.
- 2) Solar Street lighting.
- 3) Solar Pumps for agriculture.
- 4) Large Solar Power Plants

Q3)

→ Solar thermal technology uses sunlight to generate Heat, Steam, Electricity.

Solar.

Thermal Applications World wide

1) Solar Water heating.

Used in: Homes, Hotels, Hospitals, swimming pools.

Countries leading: China, USA, Germany, India.

2) Concentrated Solar Power Plants.

CSP plants generate electricity using steam turbines.

Major CSP Countries: Spain, USA, Morocco, UAE

3) Industrial Process Heat.

They require high-temperature heat for:

Food Processing, Textile industries, Chemical plant,

Dairy operations.

4) Solar Desalination

Used to convert seawater into fresh water

Used in Countries like Middle East region &

Coastal dry regions.

5) Solar Cooking Systems

They cook food using sunlight converting it into heat.

Ex: Trimala Temple Solar kitchen.

6) Solar Space Heating

Used in colder Countries for Room heating etc.

Common in Europe & Canada.